

Function	Source	Status	Key number (recomputed)
Feeding decision after touchdown (MN9 above 10 Hz)	● BRAIN	● shown (given the hand-made trigger)	MN9 > 10 Hz at touchdown: 6/6 seeds, 5/8 starts
Trigger of the feeding decision (leg contact drives labellar sugar GRNs)	□ HAND-MADE	● shown (hand-made)	contact → 36 labellar GRNs (not leg GRNs)
Leg GRN → MN9 path (the natural trigger)	● BRAIN	✗ not shown	MN9 0.0 Hz in 3/3 seeds (notebook)
Route to the target	□ HAND-MADE	● shown (hand-made)	n1 S1ΛS2 6/6 seeds, 8/8 starts; no brain share
Landing sequence (phase machine: take-off, cruise, approach, descend, touchdown)	□ HAND-MADE	● shown (hand-made)	touchdown 6/6 n1 seeds
Attitude stabilisation (halter PD) and head reflex	▲ REFLEX	● shown (hand-set)	hand-set gains; neck not connected to the brain
Brain-only flight (hand-made flight programme only)	● BRAIN	✗ not shown	n2 S1ΛS2 0/6, touchdown 0/6
Steering from brain DNs (DNp15, closed loop)	● BRAIN	✗ not shown	validation: optomotor sign 12/12, b1 6/6, b2 3/6, b3 3/6
Altitude and thrust from DNg02	● BRAIN	✗ not shown	DNg02 spikes in 7 runs: 0
Olfaction (brain's olfactory circuit)	● BRAIN	✗ not shown	0 of 8 rates return to rest (O1); N1/N2 and upstream-style drive: none
Vision → descending neurons (open loop)	● BRAIN	✗ not shown	1/21 a-priori tests passed (DNp15)
Loom tracking by DNs (open loop)	● BRAIN	✗ not shown	0/20 a-priori tests passed
Trained readout of the DNs for the route commands (T1)	✚ TRAINED	☒ not completed	S1 0/4 in all three arms; trials 2–4 not run
Visual input to the brain	◇ FLYVIS	● shown (input only)	32 boundary-layer types, 34,121 neurons

● shown

 ✗ not shown

 ☒ not completed

Words in parentheses give the limit of a shown result (hand-made trigger, hand-set constants).